

Sparse and Deep Gaussian Process Closure Modeling for Non-Stationary Two-Dimensional β -Plane Vorticity Flows Past Idealized Obstacles

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High-resolution simulations that fully resolve all spatio-temporal scales of geophysical and turbulent flows remain a challenge in large ocean domains [1, 2]. Large-eddy simulations (LES) make these computations tractable by filtering out subgrid-scale (SGS) features, but require accurate closures to remain stable and faithful; without them, solutions can drift, lose energy at the wrong rate, or develop spurious coastal artifacts [3–5]. Classical analytical closures based on the eddy-viscosity hypothesis, such as the Smagorinsky and Leith models and their dynamic variants [6–8], were developed primarily for three-dimensional homogeneous turbulence. A recent benchmarking study [9] has shown that they logically only weakly capture the SGS forcing in two-dimensional vorticity flows in the presence of coastal boundaries and interior landforms, motivating the development of data-driven closures. Among such approaches, neural-network closures [10–15] have shown promise but typically return only a deterministic point estimate of the SGS term, while the mapping from resolved to unresolved scales is fundamentally non-invertible and the closure is therefore intrinsically stochastic [16–21]. This non-uniqueness becomes especially pronounced in non-stationary flows, where the wake statistics themselves drift in time and a single deterministic correction can likely not represent the spread of admissible SGS responses.

In this work, we develop and evaluate sparse and deep Gaussian process (GP) closures for under-resolved, non-stationary two-dimensional classical and β -plane vorticity flows past idealized obstacles, where non-stationarity is driven by a time-modulated inflow velocity $U_\infty(t)$ that produces a continuously evolving wake, with shedding frequency, wake width, and subgrid-scale statistics all drifting along the trajectory. Gaussian processes are well suited to closure modeling in fluids: they encode smoothness and invariance through kernels, learn nonparametric mappings from data, and return calibrated posterior uncertainty alongside the mean correction [22]. To model the SGS forcing, we adopt a hybrid architecture in which a shallow convolutional neural network (CNN) extracts local features from the resolved fields, motivated by the finding that the SGS stresses in two-dimensional and β -plane turbulence depend only weakly non-locally on the coarse fields and are therefore well represented by small CNNs [20, 23, 24]; the extracted features are then mapped to the closure source term by a sparse variational GP head with inducing points [25, 26]. GP regression has been used directly as the predictive

component of data-driven turbulence closures: Ho et al. [27] learn an ensemble of per-case GP emulators that map local flow features to a field-inversion-derived correction for the k - ω SST model, and exploit the ensemble’s predictive variance to improve generalization to unseen geometries; Zighed et al. [28] use GP regression for super-resolution, recovering fine-resolution fields from filtered ones in a hybrid VAE-transformer-GP framework for Kolmogorov flow. Here the GP head plays a different role: it directly predicts the SGS closure source term in the filtered vorticity equation from CNN-extracted resolved-field features, with the deep-kernel-learning construction [29] combining the representational efficiency of shallow CNNs for such flows with the predictive accuracy and calibrated posterior uncertainty of GP regression. The closure is embedded intrusively in the pseudo-spectral solver of [9] and evaluated online at every coarse step, keeping it consistent with the numerics and providing a per-step uncertainty estimate. Rather than training on a single trajectory, we train on an *ensemble* of fully resolved realizations driven by different modulation functions $U_\infty(t)$ (varying in amplitude, frequency, phase, and waveform), so that the GP posterior naturally encodes the spread of SGS responses consistent with a family of non-stationary inflow forcings.

We consider the two-dimensional vorticity-streamfunction (ω, ψ) formulation on the β -plane,

$$\frac{\partial \omega}{\partial t} + J(\psi, \omega) = \nu \nabla^2 \omega - \mu \omega - \beta \frac{\partial \psi}{\partial x} + F + F_{\text{obs}}, \quad (1)$$

$$(u, v) = \left(-\frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial x} \right), \quad \omega = \nabla^2 \psi,$$

where $J(\psi, \omega)$ is the nonlinear Jacobian, ν the kinematic viscosity, μ the linear bottom-drag coefficient, and β the meridional gradient of the Coriolis parameter. The planetary-vorticity-gradient term $\beta \partial_x \psi$ supports Rossby waves and zonal jets and provides the geophysical character of the flow. Idealized obstacles (circular islands, coastal capes, and submarine-like bluff bodies) are imposed without mesh modification through a Brinkman volume-penalization forcing F_{obs} [30, 31], and the inflow velocity upstream is prescribed as a time-dependent signal $U_\infty(t)$. The boundary-generated wakes are inherently non-stationary, with transient vortex shedding.

We assess the proposed sparse and deep GP closures on three obstacle-induced transient wake regimes: (i) flow past an idealized circular island, (ii) flow past a coastal cape with several β ’s, and (iii) flow past an idealized submarine. For each obstacle, we generate an ensemble of fully resolved reference simulations driven by a family of modulation functions $\{U_\infty^{(i)}(t)\}_{i=1}^{N_{\text{ens}}}$ that vary in amplitude, frequency, phase, and waveform. Models are trained on filtered downsamplings of a

subset of these realizations and evaluated online at coarser resolutions on *held-out modulation functions* that were not seen during training, providing a clean test of out-of-distribution generalization across non-stationary forcing regimes within the same physical setup. Performance is quantified using a comprehensive suite of skill metrics: pointwise L_2 and L_∞ errors of the resolved fields, *a-priori* and *a-posteriori* Pearson correlation of the SGS forcing Π and the vorticity ω with their filtered references [9], total kinetic-energy and enstrophy spectra, time-resolved wake-integrated lift and drag coefficients $C_L(t)$, $C_D(t)$, and instantaneous Strouhal number $St(t)$. We additionally evaluate the calibration of the GP posterior uncertainty by comparing predictive intervals against the empirical SGS error along each trajectory, and assess whether the posterior widens appropriately on out-of-distribution modulation regimes.

REFERENCES

- [1] S. B. Pope, “Turbulent flows,” 2000.
- [2] B. Fox-Kemper, A. Adcroft, C. W. Böning, E. P. Chassignet, E. Curchitser, G. Danabasoglu, C. Eden, M. H. England, R. Gerdes, R. J. Greatbatch *et al.*, “Challenges and prospects in ocean circulation models,” *Frontiers in Marine Science*, vol. 6, p. 65, 2019.
- [3] Y. Zhiyin, “Large-eddy simulation: Past, present and the future,” *Chinese journal of Aeronautics*, vol. 28, no. 1, pp. 11–24, 2015.
- [4] P. F. J. Lermusiaux, “Numerical fluid mechanics,” MIT OpenCourseWare, May 2015. [Online]. Available: <https://ocw.mit.edu/courses/mechanical-engineering/2-29-numerical-fluid-mechanics-spring-2015/lecture-notes-and-references/>
- [5] C. Zhang, P. Perezhogin, A. Adcroft, and L. Zanna, “Addressing out-of-sample issues in multi-layer convolutional neural-network parameterization of mesoscale eddies applied near coastlines,” *Journal of Advances in Modeling Earth Systems*, vol. 17, no. 5, p. e2024MS004819, 2025.
- [6] J. Smagorinsky, “General circulation experiments with the primitive equations: I. the basic experiment,” *Monthly weather review*, vol. 91, no. 3, pp. 99–164, 1963.
- [7] C. Leith, “Stochastic models of chaotic systems,” *Physica D: Nonlinear Phenomena*, vol. 98, no. 2-4, pp. 481–491, 1996.
- [8] M. Germano, U. Piomelli, P. Moin, and W. H. Cabot, “A dynamic subgrid-scale eddy viscosity model,” *Physics of Fluids A: Fluid Dynamics*, vol. 3, no. 7, pp. 1760–1765, 1991.
- [9] A. N. Suresh Babu, A. Sadam, and P. F. J. Lermusiaux, “Evaluation of analytical turbulence closures for quasi-geostrophic ocean flows with coastal boundaries,” in *OCEANS 2025 IEEE*. IEEE, 2025, arXiv:2508.06678.
- [10] A. Ross, Z. Li, P. Perezhogin, C. Fernandez-Granda, and L. Zanna, “Benchmarking of machine learning ocean subgrid parameterizations in an idealized model,” *Journal of Advances in Modeling Earth Systems*, vol. 15, no. 1, 2023.
- [11] A. Gupta and P. F. J. Lermusiaux, “Neural closure models for dynamical systems,” *Proceedings of The Royal Society A*, vol. 477, no. 2252, pp. 1–29, Aug. 2021.
- [12] —, “Generalized neural closure models with interpretability,” *Scientific Reports*, vol. 13, p. 10634, Jun. 2023.
- [13] Y. Guan, A. Chattopadhyay, A. Subel, and P. Hassanzadeh, “Stable a posteriori les of 2d turbulence using convolutional neural networks: Backscattering analysis and generalization to higher re via transfer learning,” *Journal of Computational Physics*, vol. 458, p. 111090, 2022.
- [14] H. Frezat, J. Le Sommer, R. Fablet, G. Balarac, and R. Lguensat, “A posteriori learning for quasi-geostrophic turbulence parametrization,” *Journal of Advances in Modeling Earth Systems*, vol. 14, no. 11, p. e2022MS003124, 2022.
- [15] K. Srinivasan, M. D. Chekroun, and J. C. McWilliams, “Turbulence closure with small, local neural networks: Forced two-dimensional and β -plane flows,” *Journal of Advances in Modeling Earth Systems*, vol. 16, no. 4, p. e2023MS003795, 2024.
- [16] J. Berner, U. Achatz, L. Batté, L. Bengtsson, A. de la Cámara, H. M. Christensen, M. Colangeli, D. R. B. Coleman, D. Crommelin, S. I. Dolaptchiev *et al.*, “Stochastic parameterization: Toward a new view of weather and climate models,” *Bulletin of the American Meteorological Society*, vol. 98, no. 3, pp. 565–588, 2017.
- [17] T. N. Palmer, “Stochastic weather and climate models,” *Nature Reviews Physics*, vol. 1, no. 7, pp. 463–471, 2019, also discussed in QJRMS reviews.
- [18] W. Edeling and D. Crommelin, “Reducing data-driven dynamical subgrid scale models by physical constraints,” *Computers & Fluids*, vol. 201, p. 104470, 2020.
- [19] A. P. Guillaumin and L. Zanna, “Stochastic-deep learning parameterization of ocean momentum forcing,” *Journal of Advances in Modeling Earth Systems*, vol. 13, no. 9, p. e2021MS002534, 2021.
- [20] P. Perezhogin, L. Zanna, and C. Fernandez-Granda, “Generative data-driven approaches for stochastic subgrid parameterizations in an idealized ocean model,” *Journal of Advances in Modeling Earth Systems*, vol. 15, no. 10, p. e2023MS003681, 2023.
- [21] E. Williams and D. Darmofal, “Stochastic generative methods for stable and accurate closure modeling of chaotic dynamical systems,” *arXiv preprint arXiv:2504.09750*, 2025.
- [22] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [23] K. Srinivasan, M. D. Chekroun, and J. C. McWilliams, “Turbulence closure with small, local neural networks: Forced two-dimensional and β -plane flows,” *Journal of Advances in Modeling Earth Systems*, vol. 16, no. 4, p. e2023MS003795, 2024.
- [24] A. Ross, Z. Li, P. Perezhogin, C. Fernandez-Granda, and L. Zanna, “Benchmarking of machine learning ocean subgrid parameterizations in an idealized model,” *Journal of Advances in Modeling Earth Systems*, vol. 15, no. 1, p. e2022MS003258, 2023.
- [25] M. Titsias, “Variational learning of inducing variables in sparse gaussian processes,” in *Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2009, pp. 567–574.
- [26] J. Hensman, N. Fusi, and N. D. Lawrence, “Gaussian processes for big data,” in *Proceedings of the 29th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2013, pp. 282–290.
- [27] J. Ho, N. Pepper, and T. Dodwell, “Probabilistic machine learning to improve generalisation of data-driven turbulence modelling,” *arXiv preprint arXiv:2301.09443*, 2023.
- [28] I. Zighed *et al.*, “Leveraging scale separation and stochastic closure for data-driven prediction of chaotic dynamics,” *Data-Centric Engineering*, 2025, arXiv:2510.24583.
- [29] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing, “Deep kernel learning,” in *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2016, pp. 370–378.
- [30] P. Angot, C.-H. Bruneau, and P. Fabrie, “A penalization method to take into account obstacles in incompressible viscous flows,” *Numerische Mathematik*, vol. 81, no. 4, pp. 497–520, 1999.
- [31] K. Schneider and M. Farge, “Numerical simulation of the transient flow behaviour in tube bundles using a volume penalization method,” *Journal of Fluids and Structures*, vol. 20, no. 4, pp. 555–566, 2005.